

# PYQ-Calibrated Final Retrieval and Audit — Detailed Lecture Notes

JKSSB Computer Awareness

October 8, 2026

## 58.1 What this capstone is

This lesson is the exam-eve mixed-retrieval capstone for the JKSSB Computer Awareness course. It does not introduce new syllabus facts. It teaches the candidate how to attack each JKSSB item format, how to audit negative stems, how to handle defective options, and how to write an answer rationale that names the trap in every distractor. The accompanying drill set holds the course depth ratio: 50% L1 direct recall, 35% L2 contrast and classification, 15% L3 one-step application. All worked mini-examples below are original items (*[Original]*) built on concepts from the Real-PYQ Evidence Registry and the trap registry; none reproduces an official paper item verbatim.

**Definition (Retrieval at exam depth).** *Answering a mixed set spanning all units under the real depth mix and the real penalty (minus one-quarter mark per wrong answer, blank scoring zero), using a fixed per-format attack and a written rationale for every item.*

| Term                 | Meaning in this lesson                                         |
|----------------------|----------------------------------------------------------------|
| Negative stem        | A stem containing NOT, EXCEPT, incorrect, or CANNOT.           |
| Near-neighbour       | A distractor formed by swapping one word of a confusable pair. |
| Statement-evaluation | An item with statements I/II/III and combination options.      |
| Defective option     | Official option wording too loose to stand as fact.            |
| Rationale            | Why the key is right plus why each distractor is wrong.        |

## 58.2 Format attacks

### NOT/EXCEPT items

Circle NOT, EXCEPT, or incorrect before reading the options. Mark each option True or False in the margin and pick the odd one out. The three true options are deliberately attractive, so never select the first true statement seen. The worked pattern, after PYQ-03 (*[Original]*): “Which is NOT true of cache?” with options on recent data, speed, backup, and locality. The backup option is the single false one, because cache holds copies of data that live in RAM and is not a backup.

### Near-neighbour items

Retrieve the triple for every key term: what it is, what it is not, and the nearest confusion. The productive pairs are RAM vs ROM, POP3 vs IMAP, Section Break vs Page Break, digital signature vs encryption, MICR vs OCR vs OMR, and compiler vs interpreter. The pattern after TRAP-15

(*[Original]*): “A digital signature encrypts the mail body so only the recipient can read it” is wrong, because a signature provides authenticity and integrity while encryption provides confidentiality. One swapped verb flips the item.

### Abbreviation and shortcut items

Say the full form on first sight: POP3 is Post Office Protocol version 3, ISP is Internet Service Provider, MICR is Magnetic Ink Character Recognition. For shortcuts, keep the exact key: F5 starts the slide show from the beginning, F7 runs spell check, and an Excel formula must start with the equals sign. Single-letter swaps (IMAP vs a coined variant, POP3 vs SMTP) are the standard distractors, so spell each abbreviation out rather than recognising its shape.

### Statement-evaluation items

Judge each statement alone, writing T or F beside I, II, and III, and only then keep the option whose combination matches. The pattern after PYQ-03 and PYQ-04 (*[Original]*): I. Registers are the fastest storage (true). II. Cache is faster than main memory (true). III. RAM is non-volatile (false). The surviving combination is I and II only. The false statement is usually the volatility claim or the Program Counter claim.

### Matching items

Lock one certain pair first and strike out every option that breaks it. The device drills decide most of these: the plotter and the sound card are output, the scanner and the light pen are input, MICR reads magnetic ink, and the BIOS initialises hardware and loads the operating system. The pattern after PYQ-08 (*[Original]*): among four device-to-role pairings, only the MICR-to-magnetic-ink pairing survives. When two options survive, compare only the pairs on which they differ.

### Assertion–reason items

Judge the assertion alone, judge the reason alone, and then ask whether the reason explains the assertion. Both statements true but unlinked is the standard trap. The pattern after PYQ-18 (*[Original]*): the assertion that IPv6 uses 128-bit addresses is true, and the reason that HTTP/3 runs over QUIC rather than TCP is true, but the reason does not explain the assertion, so the “explains” option must be rejected.

**Example 58.2.1.** *A stem reads: “Which of the following is NOT an input device?” ([Original]). Step 1: the format is NOT/EXCEPT, so the answer is the single output device. Step 2: mark each option — keyboard (input, true), mouse (input, true), scanner (input, true), plotter (output, false). Step 3: the plotter is the odd one out and the answer. Step 4: re-read the stem to confirm the negative question was answered.*

## 58.3 The negative-stem audit method

1. Underline NOT, EXCEPT, incorrect, or CANNOT in the stem before reading any option.
2. Convert the stem to its positive form: find the three true statements, because the leftover wins.
3. Test every option with a written T/F mark; select the single False one.

4. Re-read the stem once more before marking the OMR to confirm the negative question was answered.

The discipline matters because the penalty is minus one-quarter mark per wrong answer while a blank scores zero. A misread NOT-stem therefore costs the mark plus the penalty. Guess only after eliminating at least two options.

## 58.4 The defective-option audit method

Some official options use wording too loose to stand as fact. The method, per AP-02, is: quote the wording verbatim, flag the defect explicitly, and teach the stable concept. Two instances recur in this course.

First, quoted verbatim: “Program Counter stores the address of the *current* instruction being executed.” Flag: the word *current* is defective. The stable concept is that the Program Counter holds the address of the *next* instruction to fetch (TRAP-04). Any option built on the “current instruction” wording must be marked false.

Second, quoted verbatim from the option wording: “BCC hides sender and recipients from *each other*.” Flag: the phrase is defective. The stable concept is that BCC hides recipients from *each other*; the sender always knows all recipients, and the To and Cc fields remain visible to all (TRAP-16). Any option claiming the sender is hidden, or that BCC works like encryption, must be marked false.

The Tier-1 final answer key always wins over any single option’s wording, so when wording and concept conflict, the concept above is what the candidate carries into the hall.

**Example 58.4.1.** *An option states “ROM allows normal read and write” ([Original], pattern after TRAP-02). Quote it, flag it: the words “and write” are defective. The stable concept is that ROM is read-only in normal operation and writing to it needs a special process. The option is false, and the rationale names the trap rather than memorising the sentence.*

## 58.5 Answer-rationale rules

Every answer carries two halves: why the key is right, and why each distractor is wrong. For each distractor, name the trap it instantiates (for example, “reverses RAM and ROM,” “swaps Section Break with Page Break,” or “states IPv6 as 256-bit”). Keep exam wording exact: abbreviations, full forms, software names, protocol names, commands, and shortcut keys verbatim, with explanatory prose in plain simple English. One fact per line. Never invent a paper number, page, or key date. Never label a generated item as a real PYQ; every item in the drill set is *[Original]*.

## 58.6 Per-unit quick-hit list

Unit 1, Fundamentals and data representation: the smallest unit of data is the bit; decimal 10 is binary 1010; ICs arrived in the third generation and microprocessors in the fourth; BCD encodes each decimal digit in 4 bits; the 1024 ladder applies unless the option says otherwise.

Unit 2, CPU and architecture: the Program Counter holds the next instruction’s address; interrupts are handled at instruction boundaries, not after the whole program; cycle counts vary by instruction; two’s-complement overflow is a wrong-sign result, not a carry out of the MSB; zero extension is for unsigned widening; scenario items ask for the stated bottleneck, not “slow CPU.”

Unit 3, Memory: RAM is volatile and ROM is non-volatile; cache is faster than main memory and holds copies, not a backup; the CPU works through main memory and never accesses secondary storage directly; the speed order is registers, cache, RAM, secondary storage.

Unit 4, Input, output, and ports: the light pen, punch-card reader, digitizer tablet, and scanner are input; the plotter, drum printer, and sound card are output; MICR reads magnetic ink, OCR reads printed text, OMR reads marks; large engineering drawings call for a plotter; the Section Break, not the Page Break, enables different headers and footers.

Unit 5, Software and operating systems: a device driver is software and the operating system is system software; the BIOS initialises hardware and loads the operating system; open-source means the source can be viewed, changed, and shared, which is distinct from freeware and shareware; multitasking runs more than one program at a time.

Unit 6, Programming: machine, assembly, and high-level languages in order of abstraction; assembler, compiler, and interpreter kept distinct; source code vs object code; the stack follows last-in-first-out order.

Unit 7, Office and file formats: an Excel formula must start with the equals sign; nested IF behaviour on blank and non-numeric input must be traced, not guessed; COUNTIFS takes range-criterion pairs while SUMPRODUCT with multiplication also counts under multiple conditions; Mail Merge combines letters with an address database; a Select query retrieves data on a condition; F5 starts the slide show from the beginning and F7 runs spell check; the extension and the format are named separately.

Unit 8, Internet, web, and email: ISP is Internet Service Provider; the at-sign separates the user name and the domain name; POP3 downloads mail to the local machine while IMAP keeps it synced on the server; secure transmission uses SSL/TLS; BCC hides recipients from each other.

Unit 9, Networking: IPv6 addresses are 128-bit with a fixed 40-byte header; TCP orders delivery while UDP does not; ICMP is a control and diagnostic protocol that transfers no application data; BGP routes between autonomous systems (inter-domain) while OSPF and RIP work inside one domain; HTTP/3 uses QUIC over UDP, not TCP; the IaaS provider manages infrastructure while the customer manages the operating system, runtime, applications, and data.

Unit 10, Cybersecurity: malware types kept distinct (virus, worm, Trojan, ransomware, spyware, adware); a digital signature gives authenticity and integrity while encryption gives confidentiality; phishing, spoofing, and social engineering are recognition items; antivirus quarantine, firewall, patching, backup, and multi-factor authentication each map to one control.

Unit 11, Governance and integration: e-NAM serves agricultural marketing; G2C, G2G, G2B, and G2E expansions are direct recall; Digital India, NeGP, e-Kranti, and NeGD roles are kept distinct; humanware, operator, and administrator roles are named exactly.

## 58.7 Exam retrieval and common confusions

Ten distinctions prevent most errors on a mixed paper. RAM is volatile and ROM is non-volatile. The Program Counter holds the next address, not the current one. Cache is faster than main memory and is not a backup. Section Break, not Page Break, changes headers and footers. A digital signature is not encryption. POP3 downloads while IMAP syncs. IPv6 is 128-bit, never 256-bit. BGP is inter-domain. HTTP/3 uses QUIC over UDP. The IaaS provider manages infrastructure, not the customer's operating system and data. Each maps to a trap in the registry, and each appears in the drill set with its distractors drawn from that trap.

## Exercises

1. Convert the stem “Which is NOT true of ROM?” into its positive audit form, and state the selection rule.
2. Quote verbatim one defective option discussed in this lesson, flag the defective word, and state the stable concept.
3. For a statement-evaluation item with statements I, II, and III, describe the order in which the statements and the options are judged.
4. For an assertion–reason item in which both statements are true but unlinked, state the correct verdict and explain why the “explains” option is wrong.
5. Write a two-half rationale for this item: “IPv6 addresses are 256-bit.” State the verdict and name the trap.

## Solutions

1. Positive form: find the three true statements about ROM; the leftover false statement wins. Mark each option True or False and select the single False one, then re-read the stem before marking.
2. Either “Program Counter stores the address of the current instruction being executed” (defective word: current; stable concept: the PC holds the next instruction’s address) or “BCC hides sender and recipients from each other” (defective phrase: sender and recipients from each other; stable concept: BCC hides recipients from each other only).
3. Judge each statement alone first, writing T or F beside I, II, and III; only then keep the option whose combination matches the T/F pattern.
4. Verdict: both statements true, but the reason is not the explanation of the assertion. The “explains” option claims a direct link that does not hold, so it must be rejected even though each statement is true.
5. Verdict: false. IPv6 addresses are 128-bit (TRAP-18); the option states the 256-bit distractor. The correction is one line: 128-bit addresses with a fixed 40-byte header.